

1. Iris Flower Classification using KNN

Dataset: Iris Dataset (UCI / Kaggle)

Dataset Link:

<https://www.kaggle.com/datasets/uciml/iris>

Tasks:

- Load the dataset using Pandas.
 - Separate features and target.
 - Split the data into training and testing sets.
 - Train a KNN classifier with K=3.
 - Predict the test set labels.
 - Calculate the accuracy score.
 - Display the Confusion Matrix.
 - Generate the Classification Report.
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2. GridSearchCV for Customer Segmentation Prediction

Dataset: Customer Personality Analysis

Dataset Link:

<https://www.kaggle.com/datasets/imakash3011/customer-personality-analysis>

Tasks:

- Create a customer response prediction problem using the Response column.
 - Handle missing values.
 - Scale the numerical features.
 - Use GridSearchCV to tune:
 - `n_neighbors`
 - `metric`
 - `weights`
 - Report the best parameters.
 - Compare tuned KNN with default KNN.
 - Evaluate using:
 - Accuracy
 - Precision
 - Recall
 - F1-score
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Optional

Credit Card Default Prediction

Dataset: Default of Credit Card Clients

Dataset Link:

<https://www.kaggle.com/datasets/uciml/default-of-credit-card-clients-dataset>

Tasks:

- Predict whether a client will default.
 - Check for class imbalance.
 - Standardize the features.
 - Train KNN with multiple K values.
 - Evaluate using:
 - Accuracy
 - Precision
 - Recall
 - F1-score
 - Determine whether accuracy alone is sufficient for this problem.
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Letter Recognition Challenge

Dataset: Letter Recognition Dataset

Dataset Link:

<https://www.kaggle.com/datasets/nishan192/letter-recognition-uci>

Tasks:

- Predict English alphabet letters (26 classes).
- Standardize the features.
- Build a baseline KNN classifier.
- Use GridSearchCV to tune:
 - `n_neighbors = [3, 5, 7, 9, 11, 15]`
 - `metric = ['euclidean', 'manhattan']`
 - `weights = ['uniform', 'distance']`
- Evaluate the final model using:
 - Accuracy
 - Confusion Matrix
 - Classification Report
- Identify the letters that are most frequently misclassified and explain possible reasons.